

AMAN SHARMA



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[linkedin Profile](#)

PERSONAL STATEMENT

Embedded Firmware Engineer with expertise in developing and optimizing firmware for microcontroller-based systems, specializing in **Vehicle Tracking Systems (VTS)** and **AIS-140 compliance**. Proficient in **Embedded C, communication protocols** (UART, SPI, I2C, CAN, TCP, MQTT, HTTP), and **flash memory management**. Experienced in designing **modular libraries, bootloader solutions, and hardware-software integration**. Strong background in **debugging, performance optimization, and mentoring junior engineers**. Adept at collaborating with cross-functional teams to drive innovation and ensure efficient system development.

KEY SKILLS

Embedded Firmware Development | USART | SPI | I2C | CAN | TIMER | ADC | STM32 | IoT | OLED | LCD | Embedded C | DMA | IAR Embedded Workbench | VTS(AIS-140) | LED Lighting System(Simpex) | Code Review Practices | Project Management | Team Building/Collaboration | Effective Communication | Problem Solving | Critical Thinking

WORK EXPERIENCE

Embedded Design Services India Pvt. Ltd. , Faridabad

Sr. Firmware Design Engineer

October 2022 – Present

(Joined as an Intern in 2022,
promoted to current role)

Technical Contribution

- Developed and optimized embedded firmware for microcontroller-based Vehicle Tracking Systems (VTS), ensuring compliance with AIS-140 and AIS-038 standards.
- Designed modular libraries to improve code reusability and maintainability.
- Implemented and optimized communication over **TCP/IP**, **HTTP**, and **MQTT** protocols.
- Worked extensively on **NAND flash memory management** for efficient data logging.
- Designed firmware using communication interfaces like **UART**, **SPI**, **I2C**, and **CAN**.
- Developed **bootloader** for firmware updates over-the-air (OTA).
- Integrated BLE modules for diagnostics and dynamic configuration.
- Conducted hardware-software integration and debugging using tools like oscilloscopes and logic analyzers.

Leadership & Collaboration:

- Mentored junior engineers in embedded development and debugging techniques.
- Collaborated with cross-functional teams including hardware, testing, and project management.
- Conducted peer code reviews and ensured adherence to embedded coding standards.

Projects & Experience:

1. Battery Management System (BMS)

- Analyzed and maintained legacy BMS firmware during training.
- Worked on safety-critical logic for battery power management.

2. RGBAW Stick with Wireless Control (RFM Remote + RFM Integration)

- Developed firmware for Simpex and Prolight products.
- Developed and maintained firmware for Prolight RGBAW Stick, including encoder logic and dynamic color control features.
- Integrated RFM wireless module to enable remote-controlled lighting operations.
- Designed and implemented a dedicated RFM-based remote control for wireless interaction with the RGBAW stick.
- Ensured reliable communication between remote and stick using the RFM module, optimizing signal integrity and response time.

3. DMS Alert & Image Extraction

- Extracted driver behavior alerts and images from DMS.
- Enabled processing and transmission of data for monitoring systems.

4. DMS + VTS Integration

- Combined DMS with VTS to monitor driver behavior.
- Enabled real-time alert sending to TCP server and image upload to HTTP server.

5. Vehicle Tracking System (VTS) Projects

- Developed firmware for AIS140 and AIS038-compliant VTS for clients like TML, IOCL, AEPL, Nippon, and JIO.
- Implemented real-time alerts (ignition, over-speed, panic), GPS data logging, and vehicle diagnostics.
- Used **Quectel EC200** 4G LTE module for communication (TCP, HTTP, MQTT).
- Integrated **BLE** for local configuration and diagnostics.
- Maintained modular firmware architecture to support multiple customer configurations and protocols.

6. Load Dump Generator

- Designed firmware to test PCBs for high-voltage surge (100V for 10ms) compliance.

7. Low Speed Warning System(LSWS) for EV Buses

- Developed continuous virtual motor sound system for Electric Buses to enhance pedestrian awareness at low speeds.
- Implemented logic to play synthetic engine noise when the vehicle is in motion, complying with EV acoustic vehicle alert requirements.

8. Battery Level Indicator

- Implemented logic to measure and display battery levels accurately in embedded systems.

9. BLDC Motor Control

- Worked on firmware-level feature additions and modifications for an existing BLDC motor control system.
- Implemented new functional requirements and configuration updates based on limited access to source code.
- Developed driver code for BLDC regulators.

EDUCATION

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| ● Master's Degree (2023-2025) | Amity University, Noida Pursuing |
| ● Bachelor's Degree (2019-2023) | Satyug Darshan Institute of engineering & technology (B.tech) Electronics and Communication Engineering. |
| ● Higher Secondary Education (2017-2019) | Govt. Model Sr. Secondary School |

LANGUAGE

Hindi : First Language

English : Upper-Intermediate

HOBBIES & INTERESTS

Playing Online Game | New technologies | Watching Anime/Movies

